Optimizing Inventory Management and Strategizing Expansion of Corindo Spices

A Proposal Report for the BDM Capstone Project

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Declaration Statement

I am working on a Project Title "**Optimizing Inventory Management and Strategizing Expansion of Corindo Spices".** I extend my appreciation to **Corindo Spices**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate:

Khushi Pandey

Name: Khushi Pandey

Date: 12/05/2025

Executive Summary

Corindo Spices is a regional wholesale spice business located in Rajbandigarh, Gouripur, West Bengal, aiming to strengthen its business and grow its operations across the state. The company offers multiple spice products and is exploring strategic improvements to support its growth.

Currently, the business is facing two major challenges. The first is an inventory management issue, mainly due to inaccurate demand forecasting and lack of seasonal insight, resulting in frequent overstocking or stockouts at critical times. The second challenge is its limited experience in entering new markets, where it must deal with the competition from established brands and understand regional customer preferences.

To address these challenges, a structured, data-driven approach is being proposed. This encompasses collecting and analysing business data including sales, stock, and order trends using tools like Microsoft Excel and Python libraries for forecasting and trend analysis. Inventory techniques such as ABC classification and time series analysis will be applied to optimize stock levels. For regional expansion, clustering techniques, SWOT analysis, and market surveys will be used to identify high-potential areas and tailor marketing strategies accordingly.

This comprehensive analytical plan aims to improve inventory accuracy and guide informed market expansion for Corindo Spices to expand confidently into new markets with a clear, evidence-based plan.

Organization Background

The company that I am working with is Corindo Spices which is a B2B enterprise, established by Bhushan Singh. Specializing in high-quality spices, the company supplies wholesalers through a well-structured 5-layer supply chain. Their manufacturing unit is located in Rajbandigarh, Gouripur, West Bengal. The company operates with a business manager (owner), a 10-member manufacturing team dedicated to maintain quality standards, and a 4-member sales team focussed on strengthening client relationships. The establishment maintains

zero employee turnover rate, with no record of resignations or layoffs since its inception, reflecting strong employee retention and a stable workforce. With a selection of 15 different spices, the company is set to introduce 3 new products in the market. in 2021 as a spice manufacturing company under the sole proprietorship of Mr. Chandra

Problem Statement

Corindo Spices faces two major problems:

1. **Inventory Management:** Corindo Spices faces difficulties in managing inventory, particularly when launching a new product. The lack of an optimised inventory system leads to inefficiencies such as stockouts or overstocking.
2. **Expansion into West Bengal state:** Despite high-quality products, the business has not been able to expand and cater to varied tastes statewide and its neighbouring regions.

Background of the Problem

- **Inventory Management:** The inventory management issue arises from the inability to accurately forecast demand which results in stock imbalances, including overstocking and stockouts, particularly during peak or off-peak seasons. A lack of real-time inventory monitoring and external factors such as fluctuating consumer demand, seasonal variations also contribute to exacerbating the management inefficiencies. Consequently, at Corindo Spices, without a well-structured inventory system, fulfilment of orders becomes inconsistent, and therefore customer satisfaction is also affected.

- **Expansion into West Bengal state:** The second major challenge at Corindo Spices involves scaling the business to new markets while also maintaining efficient supply chain logistics and a strong competitive position. Having limited experience in regional

market expansion, Corindo Spices faces competition from spice brands that are already well-established, and different customer preferences in different territories.

To overcome these challenges, Corindo Spices must adopt data-driven decision-making and a structured market entry plan to ensure long-term growth and competitiveness in the spice industry. Adopting a structured and research-based market entry strategy will help the company tailor its offerings to regional preferences, strengthen its competitive edge, and drive long-term growth in the spice industry.

Problem Solving Approach

To address the challenges of inventory management and expansion into the state of West Bengal, Corindo Spices will require a comprehensive data-driven and logical problem-solving approach. This includes a combination of methods, strategies, data collection techniques and analysis tools to ensure informed decision-making and long-term efficiency.

- **Initial Analysis:** The first step in solving any problem is to figure out the root causes of the issues. For inventory management, this would mean detecting why demand forecasting is not effective - whether it is due to erratic demand for the newly launched product, lack of seasonal trend insights, or an absence of real-time stock monitoring. For business expansion, a SWOT analysis (Strengths, Weaknesses, Opportunities, Threats) will help us assess Corindo Spices' current readiness to compete in new regional markets and cater to diverse consumer preferences. Subsequently, further steps will be implemented for detailed and in-depth analysis of the data.
- **Data Collection:** For this project, the data collected will include key operational and market indicators such as:
    - Sales data (historical and current)
    - Stock data (especially for new product)
    - Order frequency
    - Competitor product pricing in target regions

The owner of Corindo Spices has agreed to provide me the access to their internal records required for the analysis. By combining structured analysis with smart data use, the business can resolve its challenges and achieve sustainable expansion.

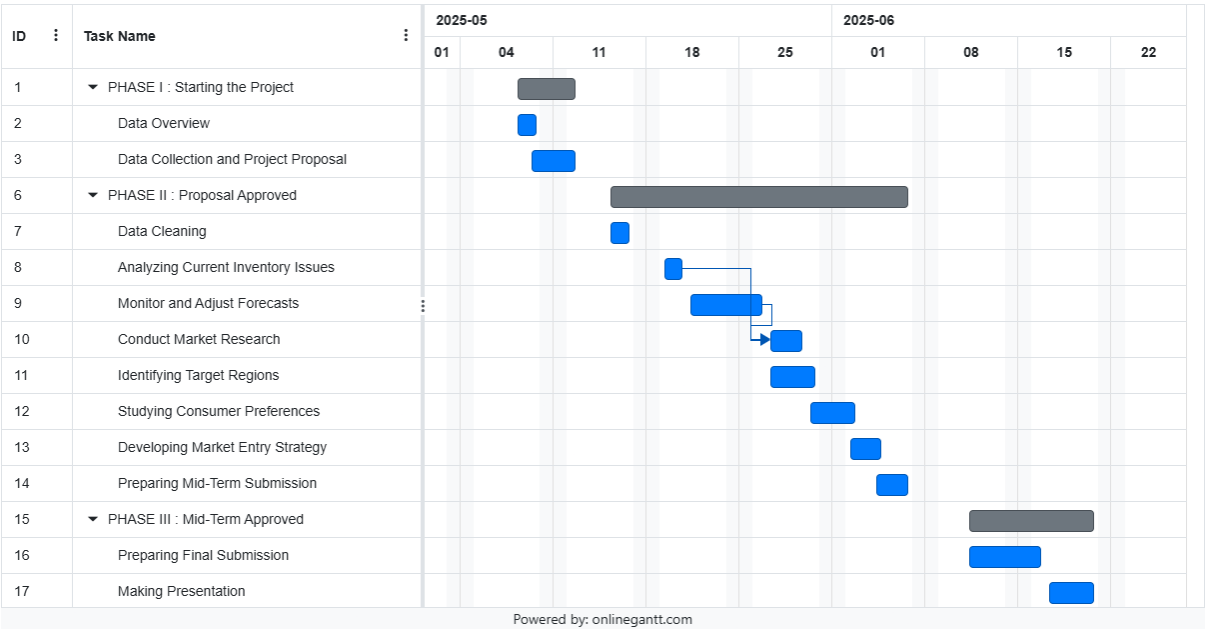
- **Tools for Analysis:** Once data is collected, it will be processed using the following tools and techniques.

  Microsoft Excel- Excel will be used for data cleaning, basic calculations and elementary analysis. For demand forecasting and inventory optimization, techniques such as time series analysis, ABC classification, and safety stock calculations can be applied.

  Python Libraries- For more complex and deeper insights, Python's libraries and frameworks such as Pandas, NumPy, Matplotlib will be used. These tools will help in better data visualisation, statistical analysis and trend identification.

  Market Expansion Analysis- To support expansion planning, clustering methods in order to segment markets on the basis of demand patterns or pricing can be utilized. Surveys and customer feedback may also help us understand local consumer preferences.

Expected Timeline



Expected timeline for completion of project.

Expected Outcome

By applying the proposed data-driven and analytical approach, Corindo Spices is expected to achieve the following outcomes:

- **Improved Inventory Management:** With the application of statistical analysis and calculations, the company will be able to forecast demand more accurately and maintain optimal inventory levels. This will reduce issues like overstocking and stockouts, especially during peak and off-peak seasons.
- **Enhanced Decision-Making:** The use of Python tools will lead to advanced trend analysis, deeper business insights and better data visualization. This will allow the management to make informed and timely decisions especially before and after launching a new product.
- **Successful Market Expansion:** On the expansion front, through market segmentation using clustering techniques, Corindo Spices will be able to identify high-potential regions to tailor its marketing and product strategies effectively. Surveys and feedback mechanisms will also provide ground-level insights into customer expectations in new areas.